

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Computer Science Maharashtra Education Service Group A
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4646621101 Question Type : MCQ

Find the number of ways to partition 10 students into four teams $[A_1, A_2, A_3, A_4]$ so that two teams contain 3 students and two teams contain 2 students.

Options :

1. 2500
2. 6300
3. 5500
4. 1500

Question Number : 2 Question Id : 4646621102 Question Type : MCQ

A store sells $M = 4$ kind of Cookies. Find number of ways a customer can buy 15 Cookies.

Options :

1. 286
2. 646
3. 342
4. 56

Question Number : 3 Question Id : 4646621103 Question Type : MCQ

Consider the set Q of rational numbers, and let $*$ be the operation on Q defined by $a * b = a + b - ab$. What is the identity element for $*$?

Options :

1. 4
2. 0
3. 2
4. 3

Question Number : 4 Question Id : 4646621104 Question Type : MCQ

Which of the following are inverse functions ?

Options :

1. Converting Celsius to Fahrenheit
2. Currency exchange
3. Mapping raw data to binary
4. All of the above

Question Number : 5 Question Id : 4646621105 Question Type : MCQ

Which of the following are applications of propositional logic ?

- (a) Web Page Searching
- (b) System Specifications
- (c) Translating English Sentences

Options :

1. (a) and (b)
2. Only (c)
3. (a) and (c)
4. (a), (b) and (c)

Question Number : 6 Question Id : 4646621106 Question Type : MCQ

Find the duals of the Boolean expressions.

(i) $x(y + 0)$

(ii) $x \cdot 1 + (y + z)$

Options :

1. (i) $x + (y \cdot 1)$, (ii) $(x + 0)(y \cdot z)$

2. (i) $x \cdot (y \cdot 1)$, (ii) $(x \cdot 0) + (y \cdot z)$

3. (i) $x + (y + 1)$, (ii) $(x + 1) \cdot (y + z)$

4. (i) $x \cdot (y + 1)$, (ii) $(x \cdot 1) \cdot (y \cdot z)$

Question Number : 7 Question Id : 4646621107 Question Type : MCQ

In a 4 digit lottery (digit 0-9, repetition allowed), a player wins a small prize if exactly 3 digits match in the correct positions with the winning number. What is the probability of winning the small prize ?

Options :

1. $36/10000$
2. $9/10000$
3. $4/10000$
4. $12/10000$

Question Number : 8 Question Id : 4646621108 Question Type : MCQ

Consider following sets :

P : Integers less than 100

Q : Real numbers between 0 & 1

R : Positive Integers less than 1,000,000,000

S : Integers that are multiples of 7

Which of the following correctly satisfies these sets as finite (F), Countably Infinite (C) or uncountable (U) ?

Options :

1. $P \rightarrow C, Q \rightarrow U, R \rightarrow F, S \rightarrow C$
2. $P \rightarrow F, Q \rightarrow U, R \rightarrow F, S \rightarrow C$
3. $P \rightarrow C, Q \rightarrow C, R \rightarrow F, S \rightarrow F$
4. $P \rightarrow F, Q \rightarrow U, R \rightarrow C, S \rightarrow F$

Question Number : 9 Question Id : 4646621109 Question Type : MCQ

Let A denote the set of students who study Data structure, B denote set of students who study Discrete maths, C denote set of students who study C-language, D denote set of students who study Theory of computation and let E denote students who are staying in Hostel and F denote set of students who went to watch a cricket match last Monday.

Express the statement in set theoretic notation. "The student who went to see cricket match are only those who study C-language or Data structure".

Options :

1. $(E - A) - B \subseteq F$
2. $A \cap F = \phi$
3. $F \subseteq C \cup A$
4. $F \subseteq D \cap B$

Question Number : 10 Question Id : 4646621110 Question Type : MCQ

What is the 8-bit sign magnitude representation of -25 ?

Options :

1. 1001 1001
2. 1111 1001
3. 1001 1010
4. 0110 0101

Question Number : 11 Question Id : 4646621111 Question Type : MCQ

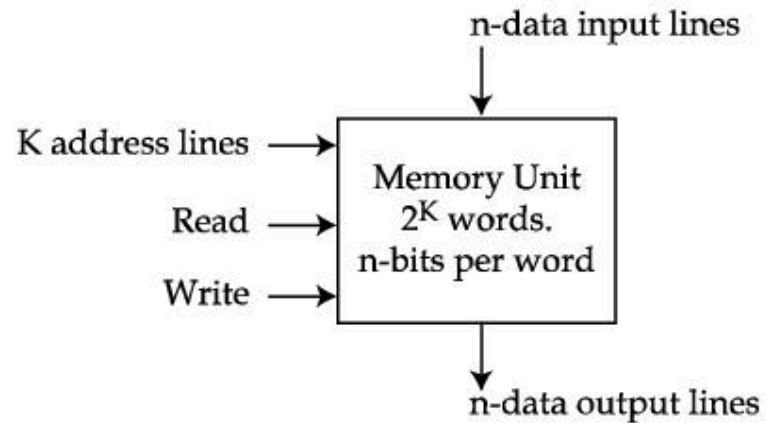
BTB stands for :

Options :

1. Buffer Transmission Bit
2. Buffer Target Bit
3. Branch Target Buffer
4. Buffer Transmission Branch

Question Number : 12 Question Id : 4646621112 Question Type : MCQ

A Random Access Memory (RAM) has $K=10$ address lines and $n=8$ data lines as shown in the diagram. What is the total memory capacity ?



Options :

1. 1 KB
2. 1 MB
3. 10 KB
4. 2 KB

Question Number : 13 Question Id : 4646621113 Question Type : MCQ

What is the 8-bit two's complement representation of -5 ?

Options :

1. 11111011
2. 00000101
3. 11111010
4. 10000101

Question Number : 14 Question Id : 4646621114 Question Type : MCQ

Which of the following best demonstrates the principle of locality in memory hierarchy design ?

Options :

1. Programs access data randomly at all times
2. Recently accessed data is likely to be accessed again
3. Secondary memory is faster than cache
4. Registers store permanent files

Question Number : 15 Question Id : 4646621115 Question Type : MCQ

DMA (Direct Memory Access) improves system performance because it :

Options :

1. Eliminates cache memory
2. Allows I/O devices to transfer data directly to/from memory without continuous CPU intervention
3. Increases processor clock frequency
4. Reduces the size of RAM

Question Number : 16 Question Id : 4646621116 Question Type : MCQ

In an instruction, the address field contains the actual operand itself, not its address. This mode is called :

Options :

1. Direct Addressing
2. Indirect Addressing
3. Register Addressing
4. Immediate Addressing

Question Number : 17 Question Id : 4646621117 Question Type : MCQ

Which of the following is an alternate name for Associative memory ?

Options :

1. Virtual Memory
2. Content Addressable Memory (CAM)
3. Random Access Memory
4. Cache Memory

Question Number : 18 Question Id : 4646621118 Question Type : MCQ

Which of the following are auxiliary memory device ?

Options :

1. RAM
2. ROM
3. Magnetic Tape
4. None of the above

Question Number : 19 Question Id : 4646621119 Question Type : MCQ

Which of the following is an example of compile time polymorphism ?

Options :

1. Operator overloading
2. Virtual function
3. Function overriding
4. Dynamic casting

Question Number : 20 Question Id : 4646621120 Question Type : MCQ

Consider the following grammar :

$\langle S \rangle \rightarrow \langle A \rangle$

$\langle A \rangle \rightarrow \langle A \rangle + \langle A \rangle$

$\langle A \rangle \rightarrow \langle A \rangle - \langle A \rangle$

$\langle A \rangle \rightarrow \langle A \rangle * \langle A \rangle / \langle A \rangle / \langle A \rangle / \langle id \rangle$

$\langle id \rangle \rightarrow a/b/c$

for given string $b + c - a / c$. The grammar is _____.

Options :

1. ambiguous
2. unambiguous
3. not valid
4. (1) & (3) both

Question Number : 21 Question Id : 4646621121 Question Type : MCQ

What does the following program print the output ?

```
#include "defs.h"

char * c [ ] = {"ENTER", "NEW", "POINT", "FIRST"};
char ** cp [ ] = {
    c + 3, c + 2, c + 1, c};
char *** cpp = cp;
main ( )
{
    printf (" % s ", ** + + cpp);
    printf (" % s ", * - - * + + cpp + 3);
    printf (" % s ", * cpp [ - 2 ] + 3);
    printf (" % s\n ", cpp [ - 1 ] [ - 1 ] + 1);
}
```

Options :

1. ENTER POINT
2. NEW FIRST
3. POINTER STEW
4. WETS RETNIOR

Question Number : 22 Question Id : 4646621122 Question Type : MCQ

For the given enum

```
enum fruit { Mango, Banana, Apple}
```

Which of the following is the correct way to declare the variable of type fruit ?

Options :

1. fruit · f1
2. f1 · fruit
3. fruit f1
4. None of the above

Question Number : 23 Question Id : 4646621123 Question Type : MCQ

Which of the following statement is **correct** ?

Options :

1. Keywords are defined by users & identifiers are predefined.
2. Keywords are predefined & identifiers are defined by user.
3. Both keywords & identifiers can be defined by users.
4. None of the above

Question Number : 24 Question Id : 4646621124 Question Type : MCQ

What does the following program print ?

```
main ()  
{  
    int x ;  
    x = -3 + 4*5 - 6 ;  
    printf ("%d\n", x);  
}
```

Options :

1. 1
2. 0
3. 11
4. -1

Question Number : 25 Question Id : 4646621125 Question Type : MCQ

The Cartesian slope-intercept equation used as the basis of line drawing algorithm is _____.

Options :

1. $x = my + b$
2. $y = mx + b$
3. $y = \frac{m}{x} + b$
4. $x^2 + y^2 = b$

Question Number : 26 Question Id : 4646621126 Question Type : MCQ

In video monitors, the ratio of vertical points to horizontal points necessary to produce equal length lines in both directions on the screen is called as _____.

Options :

1. Bit map ratio
2. Dot pitch
3. Resolution
4. Aspect ratio

Question Number : 27 Question Id : 4646621127 Question Type : MCQ

Which of the following statement is true about Raster scan display ?

Options :

1. It draws only lines and curves.
2. It stores picture as a set of line/ commands.
3. It displays images using a grid of pixels.
4. It is mainly used in oscilloscopes.

Question Number : 28 Question Id : 4646621128 Question Type : MCQ

What value can be set for the angular step size to generate a more continuous boundary on a raster display using circle Generating algorithm ?

Options :

1. $R \cos \theta$
2. $R \sin \theta$
3. $\frac{1}{r}$
4. r^2

Question Number : 29 Question Id : 4646621129 Question Type : MCQ

To generate a GLUT sphere which of the following function(s) can be used ?

Options :

1. glut wire sphere (r, n Longitudes, n Latitudes)
2. glut solid sphere (r, n Longitudes, n Latitudes)
3. Either (1) or (2)
4. None of the above

Question Number : 30 Question Id : 4646621130 Question Type : MCQ

In 2D objects, which points have a nonzero value for the winding number for Inside-Outside test ?

Options :

1. Exterior point
2. Interior point
3. A vertex
4. A boundary pixel

Question Number : 31 Question Id : 4646621131 Question Type : MCQ

Consider the query :

```
SELECT Salary AS income
```

```
FROM Employee ;
```

Which statement is true ?

Options :

1. The column name in the table changes permanently.
2. Income is a temporary alias for display.
3. A new column named income is created.
4. The query is invalid.

Question Number : 32 Question Id : 4646621132 Question Type : MCQ

A company database stores Employee ID, Employee Name, Department Name, Department Location.

The designer creates Department as an entity set instead of keeping department details as attributes of Employee. Why this design is better ?

Options :

1. It reduces normalization
2. It avoids redundancy and supports relationships
3. It increase attribute count
4. It removes the need for keys

Question Number : 33 Question Id : 4646621133 Question Type : MCQ

Which statement is TRUE about data abstraction ?

Options :

1. It exposes all internal details.
2. It focuses only on implementation.
3. It provides only necessary information to the user.
4. It avoids using methods.

Question Number : 34 Question Id : 4646621134 Question Type : MCQ

What is the degree of a table or relation with 50 rows and 5 columns ?

Options :

1. 50
2. 250
3. 5
4. 25

Question Number : 35 Question Id : 4646621135 Question Type : MCQ

A bank transfer system ensures that money is either fully transferred or not transferred at all. This property is :

Options :

1. Consistency
2. Isolation
3. Atomicity
4. Durability

Question Number : 36 Question Id : 4646621136 Question Type : MCQ

Consider the relation : R (A, B, C, D, E) Functional dependencies :

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

$D \rightarrow E$

Which is the highest normal form of relation ?

Options :

1. 1 NF
2. 2 NF
3. 3 NF
4. BCNF

Question Number : 37 Question Id : 4646621137 Question Type : MCQ

According to the formal definition, a relation schema R is in BCNF if, for every non-trivial functional dependency $X \rightarrow A$:

Options :

1. A is a prime attribute
2. X is a superkey of R
3. X is a subset of a candidate key
4. A is not a part of any key

Question Number : 38 Question Id : 4646621138 Question Type : MCQ

Consider the EMPLOYEE relation and functional dependencies (assume atomic attributes) :

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS, CHG_HOUR)

Functional dependencies :

$EMP_NUM \rightarrow \{EMP_NAME, JOB_CLASS\}$

$JOB_CLASS \rightarrow CHG_HOUR$

The EMPLOYEE relation is in :

Options :

1. Boyce-Codd Normal Form (BCNF)
2. Third Normal Form
3. First Normal Form
4. Second Normal Form

Question Number : 39 Question Id : 4646621139 Question Type : MCQ

In the ternary relationship among supplier, project and part, a tuple (S1, P1 part1) means that supplier S1 supplies part **part1** to project P1. Why can this ternary relationship NOT to be accurately replaced by three binary relationships – Supplier – Project, Project – Part and Supplier – Part ?

Options :

1. The three binary relationships cannot capture which specific supplier supplies which specific part to which specific project, they allow spurious combinations that do not exist in reality.
2. The Binary relationships require more storage space.
3. Binary relationships cannot involve entity sets that already participate in other relationships.
4. A ternary relationship is simply shorthand notation for three binary relationships and carries no additional semantic meaning.

Question Number : 40 Question Id : 4646621140 Question Type : MCQ

In a Linux Operating System, the function fork (), open () and read () are categorized as _____.

Options :

1. Compiler directives
2. Shell commands
3. System calls
4. Device drivers

Question Number : 41 Question Id : 4646621141 Question Type : MCQ

Which of the following hashing algorithm produces 32 byte result ?

Options :

1. SHA - 1
2. SHA - 512
3. SHA - 256
4. SHA - 128

Question Number : 42 Question Id : 4646621142 Question Type : MCQ

In which of the following application domains, Tiny OS is primarily used ?

Options :

1. Mobile phones
2. Automotive control system
3. Smart cards
4. Wireless sensor networks

Question Number : 43 Question Id : 4646621143 Question Type : MCQ

Consider the virtual page referencing string 10, 20, 30, 20, 40, 10, 30, 20, 40, 10 on a demand paged virtual memory system running on a computer system that has a main memory size of 3 page frames which are initially empty.

Let LRU, FIFO and optimal denote the number of page faults under the corresponding page replacement policy. Arrange the page faults in ascending order :

Options :

1. Optimal < LRU < FIFO
2. Optimal < FIFO < LRU
3. Optimal = LRU < FIFO
4. Optimal = FIFO < LRU

Question Number : 44 Question Id : 4646621144 Question Type : MCQ

The following program consist of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as

$S_0 = 1, \quad S_1 = 0, \quad S_2 = 0$

Process (P_0)	Process (P_1)	Process (P_2)
While (true) {	wait (S_1);	wait (S_2);
wait S_0 ;	Release (S_0);	Release (S_0);
print 0;		
Release S_1 ;		
Release S_2 ;		
}		

How many times will the process P_0 print '0' ?

Options :

1. At least Twice
2. Exactly Twice
3. Exactly Thrice
4. Exactly Once

Question Number : 45 Question Id : 4646621145 Question Type : MCQ

During every process switch, time taken by dispatcher to stop one process and start another running, is known as _____.

Options :

1. Content switching
2. Interrupt latency
3. Dispatch latency
4. Response time

Question Number : 46 Question Id : 4646621146 Question Type : MCQ

Consider a Resource Allocation Graph (RAG) with four processes P_1 , P_2 , P_3 & P_4 and three single instance resources R_1 , R_2 , R_3 .

The system is in the following state :

- P_1 holds R_1 and waiting for R_2
- P_2 holds R_2 and waiting for R_3
- P_3 holds R_3 and waiting for R_1
- P_4 is waiting for R_2 and currently holds no resources

Which of the following statement is most accurate for given scenario ?

Options :

1. The system is deadlock free since at least one process is not part of any cycle.
2. Deadlock exists, but only processes involved in circular wait are affected.
3. Deadlock exists among all four processes.
4. The system is in safe state as resource requests can be satisfied sequentially.

Question Number : 47 Question Id : 4646621147 Question Type : MCQ

In Round-Robin Scheduling, when a process exceeds the fixed CPU time limit, it is _____.

Options :

1. Terminated
2. Sent to I/O queue
3. Preempted and moved to the end of the scheduling list
4. Given higher priority

Question Number : 48 Question Id : 4646621148 Question Type : MCQ

What is the main performance issue associated with cache memory (L1 and L2), _____.

Options :

1. Reducing disk space
2. Ensuring high hit ratios
3. Increasing process size
4. Managing user files

Question Number : 49 Question Id : 4646621149 Question Type : MCQ

Client requirements change frequently and rapid delivery is needed. Which of the following model suits best ?

Options :

1. Waterfall Model
2. Spiral Model
3. Agile Model
4. V-Model

Question Number : 50 Question Id : 4646621150 Question Type : MCQ

Consider the basic COCOMO Model where E is the effort applied in person months, D is the development time in chronological months, KLOC is the estimated number of delivered lines of code (in thousands) and a_b , b_b , c_b , d_b have their usual meaning.

The basic COCOMO are of form :

Options :

1. $E = a_b (KLOC) \exp (b_b), D = c_b (E) \exp (d_b)$
2. $D = a_b (KLOC) \exp (b_b), E = c_b (E) \exp (d_b)$
3. $E = a_b \exp (b_b), D = c_b (KLOC) \exp (d_b)$
4. $E = a_b \exp (b_b), D = c_b (KLOC) \exp (b_b)$

Question Number : 51 Question Id : 4646621151 Question Type : MCQ

In context of modular software design, which of the following combination is desirable ?

Options :

1. High cohesion and high coupling
2. High cohesion and low coupling
3. Low cohesion and high coupling
4. Low cohesion and low coupling

Question Number : 52 Question Id : 4646621152 Question Type : MCQ

The productivity of engineers in integrating code is 2400 source statements. If there are a total of 20,000 lines of reused source code in a system and 30% of this is automatically generated. Then Number of person's effort required to integrate general code is :

Options :

1. 5 person - months
2. 4 person - months
3. 2.5 person - months
4. 3 person - months

Question Number : 53 Question Id : 4646621153 Question Type : MCQ

Standard CMMI method is used for :

Options :

1. Assessment
2. Diagnosis
3. Requirement gathering
4. Planning

Question Number : 54 Question Id : 4646621154 Question Type : MCQ

In software configuration management a basic object refers to :

Options :

1. A collection of multiple objects
2. A database of configuration
3. A testing tool
4. A unit of text, created during analysis, design, coding or testing

Question Number : 55 Question Id : 4646621155 Question Type : MCQ

Which category indicate that failure to meet the requirement would result in mission failure ?

Options :

1. Critical
2. Catastrophic
3. Marginal
4. Negligible

Question Number : 56 Question Id : 4646621156 Question Type : MCQ

In the W⁵HH principle, which question is primarily concerned with management and technical strategy for the project ?

Options :

1. Why
2. What
3. How
4. Where

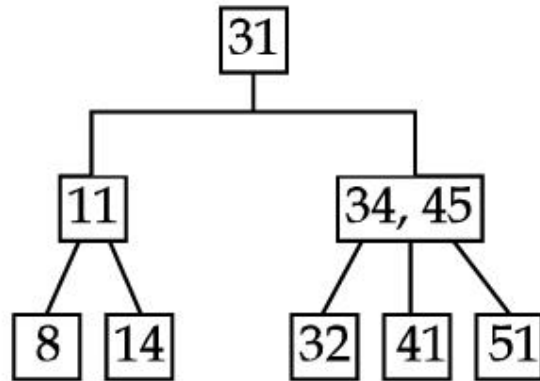
Question Number : 57 Question Id : 4646621157 Question Type : MCQ

A team lead was deciding an average operation (methods) size using Operation - Oriented Metrics. Which of the following parameters can be considered ?

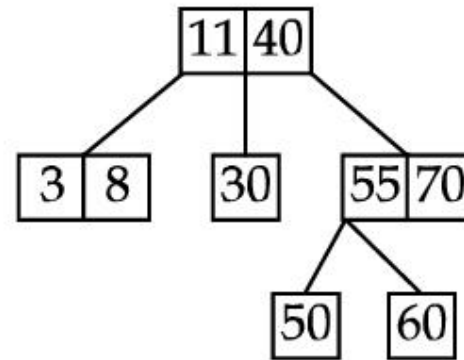
Options :

1. Number of lines
2. Number of messages sent
3. Both (1) and (2)
4. None of the above

Question Number : 58 Question Id : 4646621158 Question Type : MCQ



Dia. A



Dia. B

Which one is B tree ?

Options :

1. A
2. B
3. Both A and B
4. None of them

Question Number : 59 Question Id : 4646621159 Question Type : MCQ

According to the greedy approach for Fractional Knapsack, an optimal solution is obtained when items are chosen in :

Options :

1. Increasing order of weight
2. Decreasing order of weight
3. Increasing order of value to weight ratio
4. Decreasing order of value to weight ratio

Question Number : 60 Question Id : 4646621160 Question Type : MCQ

In the Breadth-First Search (BFS) algorithm, what data structure is used to hold the nodes that are waiting to be processed ?

Options :

1. STACK
2. QUEUE
3. Linked List
4. Binary tree

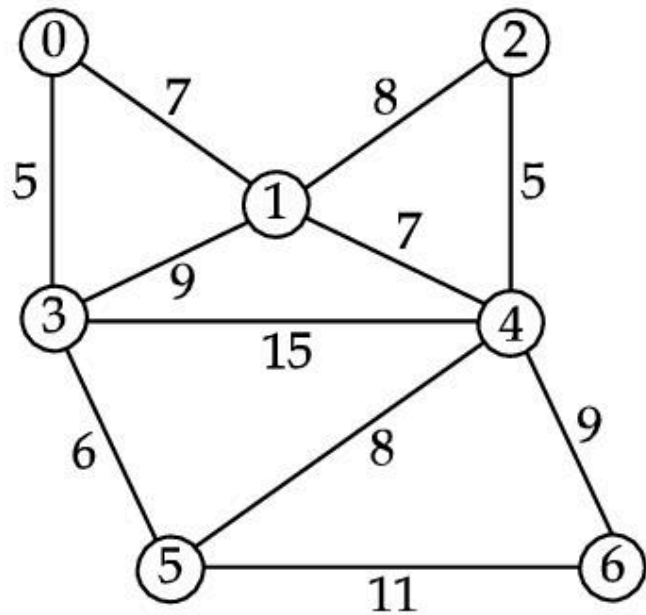
Question Number : 61 Question Id : 4646621161 Question Type : MCQ

In finding the shortest path from vertex i to j , the problem is solved by making a sequence of optimal decisions for intermediate values, which technique best models this approach ?

Options :

1. Greedy method
2. Divide and conquer
3. Back tracking
4. Dynamic programming

Question Number : 62 Question Id : 4646621162 Question Type : MCQ



Find minimum spanning tree using Kruskal's algorithm.

Options :

1. $(0, 3) \rightarrow (2, 4) \rightarrow (3, 5) \rightarrow (0, 1) \rightarrow (1, 4) \rightarrow (4, 5)$
2. $(0, 3) \rightarrow (2, 4) \rightarrow (3, 5) \rightarrow (0, 1) \rightarrow (1, 4) \rightarrow (4, 6)$
3. $(0, 3) \rightarrow (2, 4) \rightarrow (3, 5) \rightarrow (2, 1) \rightarrow (1, 4) \rightarrow (4, 6)$
4. $(1, 3) \rightarrow (2, 4) \rightarrow (3, 5) \rightarrow (0, 1) \rightarrow (1, 4) \rightarrow (4, 6)$

Question Number : 63 Question Id : 4646621163 Question Type : MCQ

Which notation is used to represent the lower bound of an algorithm ?

Options :

1. Big O notation
2. Theta notation
3. Omega notation
4. Sigma notation

Question Number : 64 Question Id : 4646621164 Question Type : MCQ

Consider the array (using 1-based indexing):

$A = \{10, 8, 2, 7, 3, 4, 9, 1, 6, 5\}$

A linear search is performed to find $VAL = 3$. The algorithm starts from the first element and compares sequentially until the value is found. How many comparisons are required, and what will be the final position (POS) of the element ?

Options :

1. 3 comparisons; POS = 4
2. 4 comparisons; POS = 5
3. 6 comparisons; POS = 6
4. 5 comparisons; POS = 5

Question Number : 65 Question Id : 4646621165 Question Type : MCQ

In an adjacency multilist of an undirected graph, each edge is stored once but linked to both vertices. If the graph has E edges, how many total links are maintained ?

Options :

1. E
2. $E + V$
3. V
4. $2E$

Question Number : 66 Question Id : 4646621166 Question Type : MCQ

The graph has 10,000 vertices and 20,000 edges. Which representation is most suitable for this graph ?

Options :

1. Adjacency Matrix
2. Adjacency List
3. Sets
4. None of them

Question Number : 67 Question Id : 4646621167 Question Type : MCQ

A binary search tree becomes right skewed when elements are inserted in _____.

Options :

1. Random order
2. Descending order
3. Ascending order
4. Alternate order

Question Number : 68 Question Id : 4646621168 Question Type : MCQ

What is the worst case time complexity for searching in a Binary search tree ?

Options :

1. $O(\log_2 n)$
2. $O(n \log n)$
3. $O(1)$
4. $O(n)$

Question Number : 69 Question Id : 4646621169 Question Type : MCQ

In a sorted array based priority queue : [841, 632, 320, 109, 43], with Rear at index 0, a new element 300 is inserted what happens to the rear pointer during this insertion ?

Options :

1. Moves forward with each shift
2. Moves backward after insertion
3. Remains fixed at index 0
4. Points to the newly inserted element

Question Number : 70 Question Id : 4646621170 Question Type : MCQ

Which of the following problem is undecidable ?

Options :

1. Determining if a given context-free grammar is ambiguous
2. Determining if a deterministic finite automata
3. Determining if a turing machine Halts on a empty Tape
4. Determining if a given regular expression produces a specific string

Question Number : 71 Question Id : 4646621171 Question Type : MCQ

Consider the string "abb cc dd eee". Each letter in the string must be assigned a binary code satisfying the following properties :

- (a) For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter.
- (b) For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter.

Among the set of all binary code assignments which satisfies the above two properties, what is the minimum length of the encoding string ?

Options :

- 1. 21
- 2. 23
- 3. 25
- 4. 30

Match the items from **List-I** and **List-II** :

List-I

- (a) $\{a^n.b^n \mid n > 0\}$ is a deterministic context free language
- (b) The complement of $\{a^n.b^n.a^n \mid n > 0\}$ is a context free language
- (c) $\{a^n.b^n.a^n\}$ is a context sensitive language
- (d) $\{a^n.b^n.c^n \mid n \geq 1\}$ is a recursive language

List-II

- (i) and a recursive language
- (ii) but not context free language
- (iii) but cannot be accepted by deterministic pushdown automata
- (iv) but not a regular

Options :

1. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
2. (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
3. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
4. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

Question Number : 73 Question Id : 4646621173 Question Type : MCQ

SLR parsing stands for _____.

Options :

1. Shift Left Reduce parsing
2. Simple LR parsing
3. Syntax LR parsing
4. None of the above

Question Number : 74 Question Id : 4646621174 Question Type : MCQ

Which of the following machine recognizes context free language ?

Options :

1. Finite Automata
2. Pushdown Automata
3. Moore Machine
4. Mealy Machine

Question Number : 75 Question Id : 4646621175 Question Type : MCQ

In Moore machine, output is associated with :

Options :

1. Previous state
2. Current state
3. Both previous and current state
4. None of the above

Question Number : 76 Question Id : 4646621176 Question Type : MCQ

What does an interior node in a syntax tree represent ?

Options :

1. A variable declaration.
2. An operation, with its children representing the arguments.
3. A token from the symbol table.
4. A memory address for code generation.

Question Number : 77 Question Id : 4646621177 Question Type : MCQ

Let the alphabet Σ be the standard 26 letters $\{a, b, c, d, \dots, z\}$. If $A = \{\text{good}, \text{bad}\}$ and $B = \{\text{boy}, \text{girl}\}$.

Consider the following three claims :

- (i) $A \cup B = \{\text{good}, \text{bad}, \text{boy}, \text{girl}\}$
- (ii) $A \cdot B = \{\text{good boy}, \text{good girl}, \text{bad boy}, \text{bad girl}\}$
- (iii) $\epsilon \in A^*$

Which of the claims are **correct** ?

Options :

1. All three : (i), (ii) and (iii) are correct.
2. Only (i) and (ii) are correct.
3. Only (i) is correct.
4. Only (ii) and (iii) are correct.

Question Number : 78 Question Id : 4646621178 Question Type : MCQ

SAT problem is :

Options :

1. P
2. NP
3. NP Complete
4. Undecidable

Question Number : 79 Question Id : 4646621179 Question Type : MCQ

If a subscriber (normally an organization) needs to provide free connections for other subscriber (normally customer). It can request the :

Options :

1. 900 Service
2. Wide-Area-Telephone Service (WATS)
3. 800 Service
4. None of the above

Question Number : 80 Question Id : 4646621180 Question Type : MCQ

Which field is used to detect errors in the IPv4 Header ?

Options :

1. Fragment Offset
2. Header Checksum
3. Destination Address
4. Type of Service

Question Number : 81 Question Id : 4646621181 Question Type : MCQ

Which type of data is described as continuous rather than discrete ?

Options :

1. Text
2. Numbers
3. Images
4. Audio

Question Number : 82 Question Id : 4646621182 Question Type : MCQ

In the OSI model, error detection and flow control are primarily the responsibility of which layer ?

Options :

1. Physical Layer
2. Data Link Layer
3. Session Layer
4. Application Layer

Question Number : 83 Question Id : 4646621183 Question Type : MCQ

Which device uses simplex mode ?

Options :

1. Telephone
2. Walkie-Talkie
3. Keyboard to CPU
4. Video Call

Question Number : 84 Question Id : 4646621184 Question Type : MCQ

Which devices work at the Data Link Layer ?

Options :

1. Router and Switch
2. Bridge and Hub
3. Bridge and Switch
4. Gateway and Router

Question Number : 85 Question Id : 4646621185 Question Type : MCQ

In a classic ethernet network two computers attempt to transmit data at the same time and their packets collide. What happens next ?

Options :

1. The switch detects the collision and retransmits both packets immediately.

Both computers lose their packets permanently and wait for the receiver to request
2. retransmission.
3. Each computer waits for a random amount of time and then tries to transmit again.

The computer with higher MAC address gets priority and retransmits first while the other
4. waits indefinitely.

Question Number : 86 Question Id : 4646621186 Question Type : MCQ

Which of the following is **not** a characteristic of Packet Switching ?

Options :

1. Efficient bandwidth utilization.
2. Packets may take different paths.
3. Variable transmission delay.
4. Dedicated communication path is established before transmission.

Question Number : 87 Question Id : 4646621187 Question Type : MCQ

Considering the sliding Window protocol. If the sender's window becomes full, then _____.

Options :

1. Receiver is shut down
2. Acknowledgement is ignored
3. Network layer is temporarily stopped until another buffer becomes free
4. Sender continue sending frames

Question Number : 88 Question Id : 4646621188 Question Type : MCQ

Imagine the agent is learning in an episodic problem. Which of the following is **true** ?

Options :

1. The number of steps in an episode is stochastic; each episode can have a different number of steps.
2. The number of steps in an episode is always the same.
3. The agent takes same action at each step during an episode.
4. None of the above

Question Number : 89 Question Id : 4646621189 Question Type : MCQ

A neural network achieves zero error on training data, but, performs poorly on unseen test data.

What is the **most appropriate** explanation ?

Options :

1. The model has low capacity.
2. The model is under fitting.
3. The model has learned spurious patterns due to limited data.
4. The learning rate is too small.

Question Number : 90 Question Id : 4646621190 Question Type : MCQ

In Hierarchical Task Network (HTN) planning, what is meant by a primitive action ?

Options :

1. An action that has defined preconditions and effects and cannot be decomposed further.
2. A high level task that can be broken down into smaller subtasks.
3. A recursive procedure used to explore the state space.
4. A stored plan retrieved from a plan library.

Question Number : 91 Question Id : 4646621191 Question Type : MCQ

What is the main limitation of using accuracy alone as a performance metric for class-imbalanced data sets ?

Options :

1. Accuracy cannot be computed without a confusion matrix.
2. Accuracy depends on prior computation of precision and recall.
3. Accuracy is only valid for large datasets (e.g. more than 10,000 instances).

4. A classifier can achieve high accuracy by correctly predicting the majority class while failing to detect the minority class.

Question Number : 92 Question Id : 4646621192 Question Type : MCQ

Which of the following is a suitable application of K-nearest neighbor (KNN) ?

Options :

1. Recommending movies based on similar users
2. Compiling a program
3. Encrypting passwords
4. Managing network Hardware

Question Number : 93 Question Id : 4646621193 Question Type : MCQ

The energy of Hopfield Network is given by

$$E = - \sum_i b_i S_i - \sum_{i < j} W_{ij} S_i S_j$$

The network has two neurons $S_1, S_2 \in \{0, 1\}$ with $W_{12} = 5$ and $b_1 = b_2 = 0$

Which state minimizes the energy ?

Options :

1. (0, 0)
2. (1, 0)
3. (0, 1)
4. (1, 1)

Question Number : 94 Question Id : 4646621194 Question Type : MCQ

Face Recognition is an application of _____.

Options :

1. Reinforcement Learning
2. Unsupervised Learning
3. Supervised Learning
4. Semi-supervised Learning

Question Number : 95 Question Id : 4646621195 Question Type : MCQ

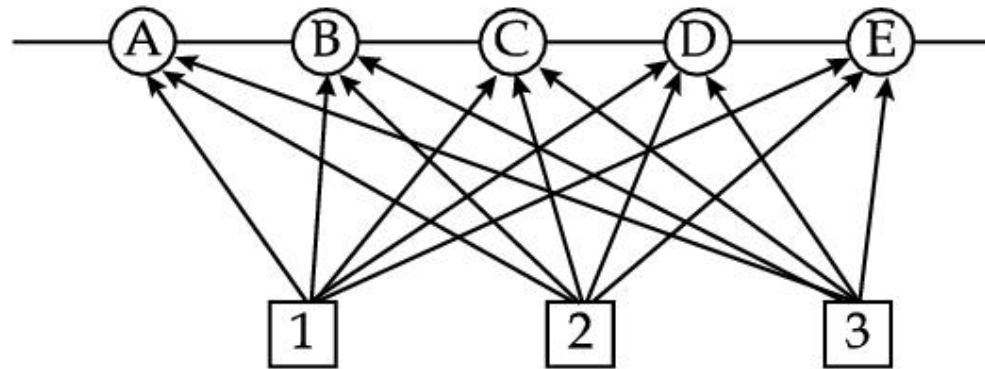
In the Self Organising Maps (SOM) algorithm, what role does the winning neuron (Best Matching Unit) play during the cooperation phase ?

Options :

1. It resets the weights of all other neurons.
2. It removes neighbouring neurons from future competition.
3. It acts as the center of a neighbourhood of neurons that are updated together.
4. It computes the output of the entire network.

Question Number : 96 Question Id : 4646621196 Question Type : MCQ

Consider the following SOM :



Find out presynaptic and postsynaptic neurons :

Options :

1. Presynaptic : A, B, C, D, E
Postsynaptic : 1, 2, 3
2. Presynaptic : A, B, C
Postsynaptic : 1, 2, 3
3. Presynaptic : 1, 2, 3
Postsynaptic : A, B, C, D, E
4. Presynaptic : 1, 2, 3
Postsynaptic : A, B

Question Number : 97 Question Id : 4646621197 Question Type : MCQ

Which of the following algorithm is called a “lazy learning” algorithm ?

Options :

1. Support Vector Machine
2. K-Nearest Neighbor
3. Hidden Markov Model
4. Naive Bayes Classifier

Question Number : 98 Question Id : 4646621198 Question Type : MCQ

Which is the first step in machine learning ?

Options :

1. Data Exploration
2. Data Acquisition
3. Model Training
4. Prediction

Question Number : 99 Question Id : 4646621199 Question Type : MCQ

In STRIPS - style planning, why are the precondition list and delete list specified separately in an operator ?

Options :

1. Because some preconditions may remain true after execution, while others may be removed.
2. Because preconditions are automatically added to the state after execution.
3. Because the delete list is applied only when goals are achieved.
4. Because preconditions replace the add list during execution.

Question Number : 100 Question Id : 4646621200 Question Type : MCQ

For neural networks, $g(\cdot)$ is a mathematical function that describes the fixing of the neurons as a response to the weighted inputs, such as the threshold function is known as :

Options :

1. Error
2. Bias
3. Activation function
4. Target